Generative AI in Multimodal Medical Image Fusion: A Comparative Analysis of GANs, VAEs, and DDPMs

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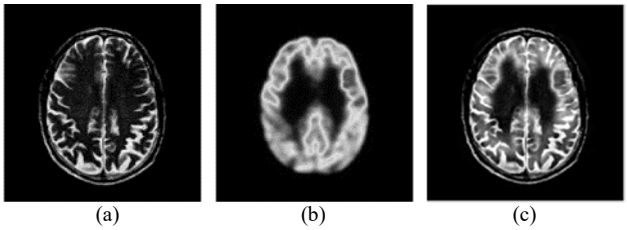
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***Abstract*— Multimodal medical image fusion (M3IF) is important for improving the medical and healthcare investigation by fusing significant information from different medical imaging modalities. Recent advancements in Generative Artificial Intelligence (Generative AI) have introduced powerful models for synthesizing high-quality fused images. This review paper presents a reasonable analysis of three significant generative AI methods—Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Denoising Diffusion Probabilistic Models (DDPMs)—for medical image fusion. Authors explained the standardized algorithmic step-by-step procedure for each method and evaluate their performance using established fusion metrics. The results in this paper highlight the strengths and weakness of each model: GANs offer photorealism, VAEs ensure robust feature integration, and DDPMs demonstrate superior detail preservation. The research paper will help to guide for the selection of correct generative models based on medical requirement and contributes to advancing AI-driven image fusion in medical imaging applications.**

Keywords—image fusion, medical image fusion, GAN, VAE, DDPM, medical images.

I. INTRODUCTION

Image fusion is one of the major research areas in medical imaging where important information from multiple modalities are fused into a single high-quality image that supports diagnosis and further medical investigation. The authors of [1] developed end-to-end unsupervised fusion framework using GAN. It provides better spatial resolution and more accurate tumor edge delineation, but this evaluation is limited to brain tumor cases. A novel end-to-end model for M3IF that can effectively fuse structural and functional information using conditional GAN (cGAN) modal [2]. It preserves both structural and functional information effectively. But it has increased training time, and higher computational cost. The authors of [3] reviewed the work of GAN in the field of M3IF. Here GAN model is categorized and analyzed based on architecture and learning methods such as Probability Distribution Distance, Dual-Discriminator GAN, Single-Discriminator GAN, Multi-Discriminator GAN, GANs based on Convolutional Neural Networks (CNN), and GANs based on Auto-Encoder structures. It provides comprehensive taxonomy of GAN models and covers a broad spectrum. But no new experimental results or fusion method is proposed. And lacks in empirical comparison or benchmarks to validate which GAN variant is superior in fusion performance. The practical example of M3IF is shown in below Fig 1.



(a) (b) (c)

Fig 1. Example of M3IF (a) MRI image, (b) SPECT image (c) Fused image

The authors of [4] developed M3IF method that performs well even with small-scale datasets to report the challenge of limited publicly accessible medical image datasets by using knowledge distillation and explainable AI in a dual-discriminator GAN framework. It uses Knowledge Distillation + Explainable AI + Dual Discriminator GAN. It provides efficient training with small datasets, reduced overfitting and improves interpretability. But can be unstable and require careful tuning during training. The authors in [5] explored the prospective of multimodal generative AI (described to as GenMI) in automating medical image interpretation. In terms of radiology, the progress, and challenges in using GenMI are discussed. It provides automation of reporting, has high potential accuracy, and supports clinical decision-making. It has difficulty in rigorously validating the correctness of generated reports, and lacks explainability and interpretability of complex generative models. The major contribution of this survey paper includes:

(a) This survey paper presents a detailed comparative study of three leading generative models—GANs, VAEs, and DDPMs—for the task of M3IF. Each method is analyzed in terms of its architecture, fusion strategy, loss design, and performance on standard fusion benchmarks.

(b) This paper formalized and presented standardized, step-by-step algorithmic pipelines for implementing GAN-based, VAE-based, and DDPM-based fusion methods.

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(c) The paper provides critical insights into the trade-offs between generation quality, structural preservation, and modality retention for each model class.

After the introduction, Section 2 describes the GAN-based, VAE-based, and DDPM-based fusion methods with an algorithm. Section 3 discuss the comparative results, and the paper is concluded in Section 4.

II. METHODOLOGY

Input: Input medical images from different modalities (e.g., MRI and CT), denoted as I_1, I_2, ..., I_n

Output: Fused medical image I_f

A. M3IF Using GANs

The step-by-step procedure of M3IF Using GANs is also explained below:

Input: Pre-trained or end-to-end trainable GAN model

Step 1: Preprocessing

1.1. Register the source images to ensure spatial alignment.

1.2. Normalize the intensity values of the input images to a common range.

Step 2: Feature Extraction

2.1. Use an encoder (CNN-based or transformer-based) to extract modality-specific features: $F_i = E(I_i)$, where E is the feature extractor.

2.2. Optionally, apply attention mechanisms to enhance salient features from each modality.

Step 3: Feature Fusion

3.1. Combine the extracted features F_1, F_2, ..., F_n using a fusion strategy:

- Simple concatenation or addition
- Adaptive fusion via attention weights
- Learnable fusion through fully connected layers

Resulting in a fused feature map: F_f

Step 4: Image Generation

4.1. Use the generator network G to synthesize the fused image from fused features: I_f = G(F_f)

Step 5: Adversarial Learning

5.1. Introduce a discriminator network D, which attempts to distinguish the fused image I_f from real (or high-quality reference) images.

5.2. Train G and D in an adversarial setting using the following loss functions:

- Adversarial Loss:
  L_adv = E[log D(I_real)] + E[log(1 - D(I_f))]
- Content Loss:
  L_content = ||I_f - I_target||
- Fusion Loss:
  Custom-designed based on energy, structure, or gradient retention.

Step 6: Optimization

6.1. Minimize the total loss to update the generator:

$G^* = \arg\min_G \max_D (L_{adv} + \lambda_1 L_{content} + \lambda_2 L_{fusion})$

6.2. Update the discriminator D to better distinguish fused from real/reference images.

Step 7: Evaluation

7.1. Assess the quality of the fused image using different objective metrics.

7.2. Optionally, validate the fusion outcome through expert radiologist feedback or downstream tasks (e.g., segmentation or diagnosis).

B. M3IF Using VAEs

Step 1: Preprocessing

1.1. Perform image registration to align the source modalities spatially.

1.2. Normalize the intensity values of each modality to a standard range (e.g., [0, 1] or [-1, 1]).

Step 2: Encoding with VAEs

2.1. For each source image I_i, use a modality-specific encoder E_i to map the image to a latent space: $z_i \sim q_i(z|I_i)$.

2.2. Extract mean μ_i and variance σ_i^2 to define the latent distribution.

2.3. Sample using reparameterization: $z_i = \mu_i + \sigma_i * \varepsilon$, where $\varepsilon \sim N(0, I)$.

Step 3: Latent Space Fusion

3.1. Fuse the sampled latent vectors z_1, z_2, ..., z_n using a strategy such as:

- Weighted averaging: $z_f = \sum w_i * z_i$
- Concatenation followed by transformation
- Attention-guided fusion

Resulting in a fused latent vector z_f.

Step 4: Decoding

4.1. Use a shared decoder D to reconstruct the fused image: I_f = D(z_f).

Step 5: Loss Function Design

5.1. VAE reconstruction loss: $L_{rec} = \sum ||I_i - \hat{I}_i||^2$

5.2. KL divergence loss: $L_{KL} = \sum D_{KL}(q_i(z|I_i) \,||\, p(z))$, where $p(z) \sim N(0, I)$

5.3. Optional fusion consistency loss to preserve complementary features.

Step 6: Optimization

6.1. Minimize the total loss: $L_{total} = L_{rec} + \beta * L_{KL} + \lambda * L_{fusion}$

where β and λ are hyperparameters.

Step 7: Evaluation

7.1. Evaluate I_f using various objective metrics.

7.2. Optionally validate using downstream tasks (e.g., segmentation, diagnosis).

C. M3IF Using DDPMs

Step 1: Preprocessing

1.1. Perform image registration to ensure spatial alignment between input modalities.

1.2. Normalize image intensities to a common scale (e.g., [0, 1]).

Step 2: Feature Extraction and Fusion

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2.1. Extract modality-specific features using an encoder: $F_i = E(I_i)$.

2.2. Fuse features using strategies such as:

- Weighted average or maximum rule
- Learnable fusion via MLPs or attention
- Concatenation followed by a fusion module

Step 3: Forward Diffusion Process (Noise Addition)

3.1. Add Gaussian noise over T time steps:

$x_t = \sqrt{\alpha_t} \cdot x_{t-1} + \sqrt{(1 - \alpha_t)} \cdot \varepsilon_t$, where $\varepsilon_t \sim N(0, I)$.

3.2. Final noisy sample x_T approximates a Gaussian distribution.

Step 4: Reverse Diffusion Process (Denoising and Generation)

4.1. Train denoising model $\varepsilon_\theta(x_t, t)$ to predict noise.

4.2. Reconstruct image iteratively:

$x_{t-1} = 1/\sqrt{\alpha_t} (x_t - (1 - \alpha_t)/\sqrt{(1 - \bar{\alpha}_t)} \cdot \varepsilon_\theta(x_t, t)) + \sigma_t z$

Step 5: Loss Function

5.1. Use DDPM objective: $L_DDPM = E[\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2]$

5.2. Optionally add:

- Structural loss (SSIM)
- Gradient loss
- Modality-specific content loss

Step 6: Evaluation

7.1. Evaluate using various objective metrics.

7.2. Optionally validate via expert opinion or segmentation results.

D. Comparison of GANs, VAEs, and DDPMs

Below Table 1 compares the GANs, VAEs, and DDPMs in M3IF.

TABLE 1. COMPARISON OF GANs, VAEs, AND DDPMs

GAN	
Strengths	**Limitations**
Produce high-resolution images	Mode collapse challenge is encountered
Extract realistic textures	
VAE	
Strengths	**Limitations**
Provide a structured latent space.	Exhibit blurriness
Facilitates smooth interpolation between data points.	Sometimes obscure fine structural details
Enables robust feature integration.	
DDPM	
Strengths	**Limitations**
High fidelity and diversity	Longer training and inference times
Captures intricate anatomical structures.	

III. Results And Discussion

Fig 2, and 3 showcase datasets composed of medical images from different modalities, as referenced in [6, 7]. These figures illustrate the visual results obtained using the GANs, VAEs, and DDPMs across two separate datasets. Based on the visual analysis of the fused results from the GAN, VAE, and DDPM methods, we observe distinct differences in image quality, structure retention, and visual clarity.

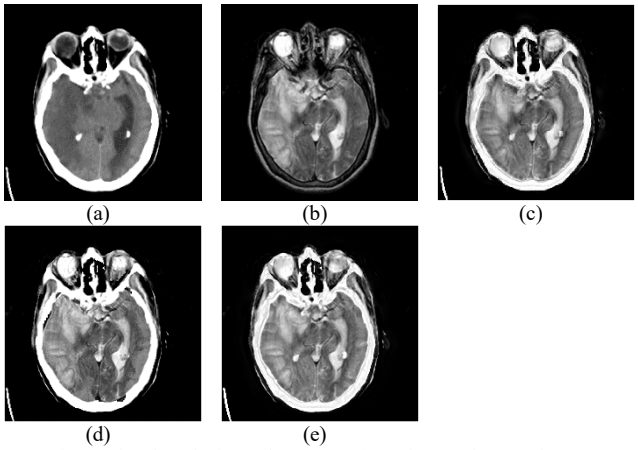
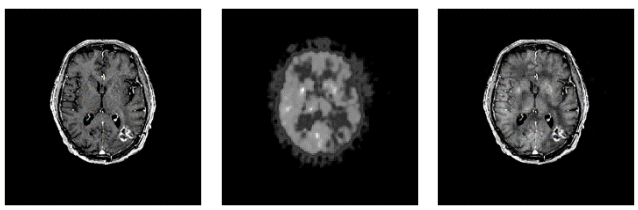


Fig 2. Visual analysis on dataset 1: (a) CT image (b) MRI image (c) GAN (d) VAE (e) DDPM

In Fig 2, The fused image produced using the DDPM method demonstrates the most visually pleasing and diagnostically relevant result. It effectively preserves the fine textures and anatomical edges, particularly in complex regions such as the orbital area and brain tissue boundaries. The image maintains a high level of sharpness, with clear transitions between different tissues, and presents minimal noise, resulting in both uniform and non-uniform areas appearing natural and well-balanced. The smoothness in homogeneous regions is well-controlled, avoiding the over-smoothing that could obscure critical clinical details.

The GAN-based image performs reasonably well, maintaining good detail in high-contrast regions and offering better edge clarity than VAE. However, subtle artifacts and moderate noise can be observed, particularly in the transitions between bone structures and soft tissues. While the texture preservation is acceptable, it lacks the finesse and realism observed in the DDPM output. The uniform areas are generally smooth, but there are occasional inconsistencies in intensity distribution, leading to a slightly less natural appearance compared to DDPM.

In contrast, the fused result obtained using the VAE model exhibits the weakest visual quality. The image appears over-smoothed, especially in non-uniform regions where fine structures are blurred or lost entirely. This leads to poor edge definition, with critical anatomical contours, such as those around the brainstem and orbital areas, appearing faded or merged. The texture details are significantly reduced, likely due to the inherent averaging effect and probabilistic reconstruction approach of VAEs. Additionally, the presence of unnatural pixelation or block-like artifacts further degrades the perceptual quality of the image, making it less suitable for diagnostic use.



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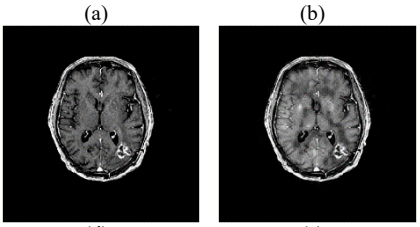


Fig 3. Visual analysis on dataset 2: (a) MRI image (b) PET image (c) GAN (d) VAE (e) DDPM

In Fig 3, The fused image generated by the DDPM model exhibits the most visually superior quality. It maintains excellent texture preservation, with finer anatomical structures clearly distinguishable. The image appears clean with minimal noise, and the smoothness in uniform regions (such as the cerebrospinal fluid areas) is well-balanced, without causing blurring in non-uniform, detail-rich regions. Additionally, edges are well-preserved, especially around the ventricular and cortical boundaries, making the fused image diagnostically valuable.

In contrast, the GAN-based fused image shows a strong capability in maintaining structural details and produces realistic contrast. However, upon close inspection, there is a slightly higher level of noise compared to DDPM. The edges are well-preserved, and textures are decently maintained, though some fine textures appear marginally distorted or over-enhanced in specific regions. Uniform regions exhibit acceptable smoothness, but the transition between homogeneous and heterogeneous areas is not as natural as in DDPM.

The VAE-generated fusion result is the least visually compelling among the three. It tends to smooth out fine textures, resulting in noticeable loss of structural details, especially in areas with high spatial variability. The uniform regions are overly smoothed, which leads to an unnatural appearance in non-uniform areas, where subtle intensity variations are suppressed. Moreover, edge sharpness is relatively poor, making it difficult to delineate certain anatomical boundaries clearly. This smoothing effect, typical of VAEs due to their probabilistic reconstruction nature, compromises the perceptual richness necessary in medical diagnostics.

In summary, the DDPM-based fusion consistently outperforms the other two approaches by generating high-fidelity images characterized by superior noise suppression, balanced smoothness in both uniform and non-uniform regions, and strong preservation of edges and textures.

In Fig 4 and 5, the visual comparisons for the fused images, their corresponding edge maps (Sobel filter), and texture maps (Local Binary Patterns - LBP) for each method: GAN, VAE, and DDPM is shown. In Fig 4, the DDPM-based edge map shows stronger and clearer edge details, indicating better edge preservation. The GAN edge and texture maps also highlight good structural features but with slightly more irregular texture distribution. The VAE outputs appear relatively weak in both edge definition and texture complexity, aligning with earlier observations of over-smoothing and detail loss.

From the visual analysis of Fig 5, DDPM again shows a more structured and continuous edge map along with high-frequency texture features, suggesting superior preservation of anatomical details. GAN maintains a good amount of structural texture but introduces more variability and slight artifacts, as seen in the texture map. VAE shows less defined edge maps and lower complexity in textures, reinforcing the earlier observation of over-smoothing and reduced structural clarity.

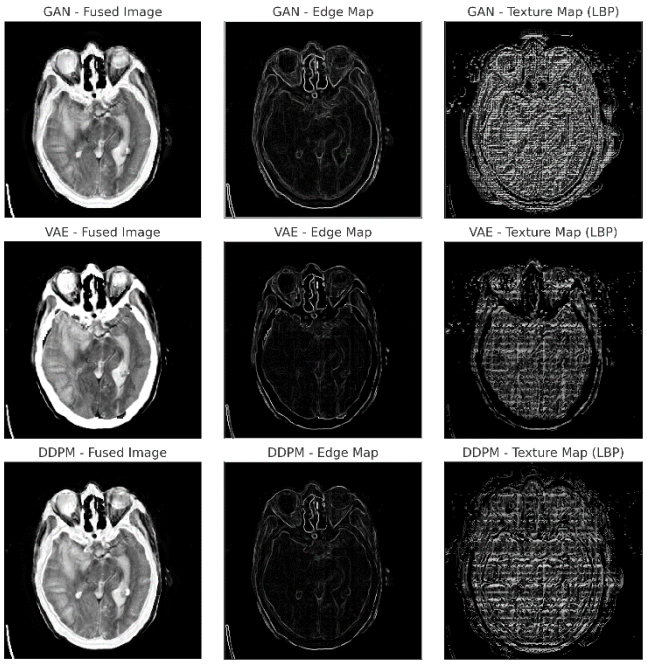


Fig 4. Fused images, edge maps, and texture maps of GANs, VAEs, and DDPMs on dataset 1

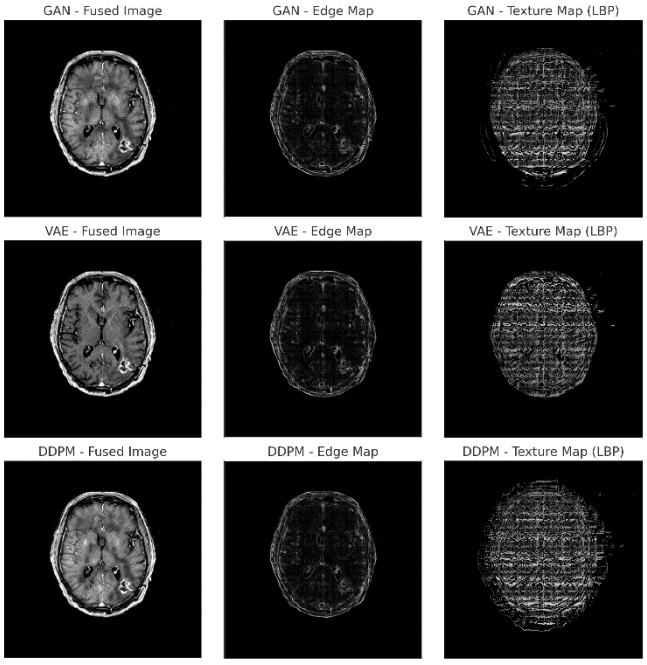


Fig 5. Fused images, edge maps, and texture maps of GANs, VAEs, and DDPMs on dataset 2

Fig 6 is enhanced radar chart that presents a normalized comparative analysis of three generative models—GANs, VAEs, and DDPMs—based on eight key fusion performance metrics: Functional Mutual Information (FMI), Fusion Symmetry (FS), Mean (M), Fusion Factor (FF), Standard Deviation (SD), Entropy (E), Fused Index (FI), and Sum of Correlation Difference (SCD). In this visualization, all

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metrics have been normalized between 0 and 1 to ensure fair comparison, with the Fusion Symmetry (FS) metric inversely normalized (since lower FS values indicate better fusion performance).

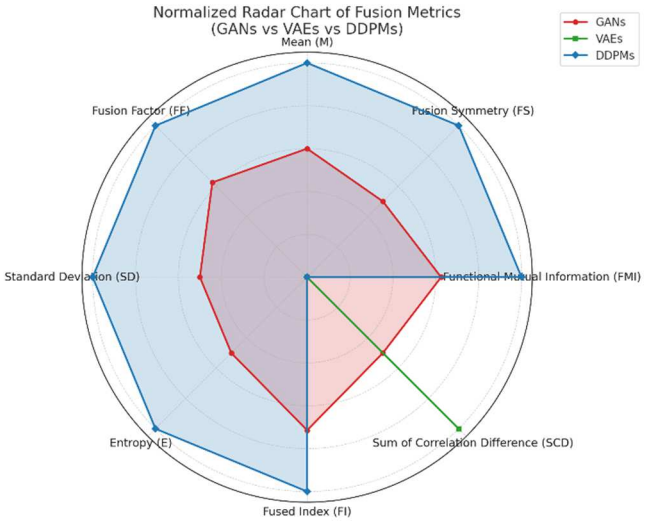


Fig 6. Normalized Radar Chart of Fusion Metrics (GANs vs VAEs vs DDPMs)

In Fig 6, DDPMs form the largest and most balanced area, indicating consistently superior performance across nearly all metrics. This includes high FMI, E, FI, and FF, signifying strong information retention, high-quality texture detail, and well-balanced fusion outputs. DDPMs also demonstrate low Fusion Symmetry, affirming better structural consistency between input and fused images. GANs perform moderately well, showing competitive values in metrics like FMI, FI, and E, but slightly lagging behind DDPMs in terms of standard deviation and texture diversity (SD and E). They still maintain a solid profile, making them reliable, though not optimal. VAEs display the smallest profile, with lower normalized scores across most metrics. This confirms earlier qualitative observations of over-smoothing, loss of edge sharpness, and weaker contrast, which result in diminished fusion performance. Although they offer a robust latent space for learning, VAEs struggle with preserving high-frequency and edge-rich details.

Overall, this figure visually confirms that DDPMs are the most effective generative approach for multimodal medical image fusion among the three, with GANs being a viable alternative, and VAEs trailing due to their inherent limitations in detail preservation. The findings presented in Fig 6 represent average outcomes derived from experiments conducted on over 50 datasets.

IV. Conclusion

In this survey paper, we presented a comprehensive comparative analysis of three prominent generative models—GANs, VAEs, and DDPMs—for the task of M3IF. Standardized algorithmic frameworks were formulated for each model, and their performance was evaluated using a suite of widely accepted fusion metrics. Quantitative results, supported by visual inspection and structural analysis through edge and texture maps, demonstrate that DDPMs consistently outperform GANs and VAEs in preserving fine anatomical details, suppressing noise, and maintaining balance in both uniform and heterogeneous regions. GANs offer competitive performance in structural fidelity and contrast preservation but exhibit minor artifacts. In contrast, VAEs underperform due to their tendency to over-smooth features, leading to loss of critical diagnostic information. The findings of this study suggest that diffusion-based generative models hold significant promise for high-quality medical image fusion tasks.

References

[1] Safari, M., Fatemi, A., & Archambault, L. (2023). MedFusionGAN: multimodal medical image fusion using an unsupervised deep generative adversarial network. BMC Medical Imaging, 23(1), 203.

[2] Le, Z., Huang, J., Fan, F., Tian, X., & Ma, J. (2020, October). A generative adversarial network for medical image fusion. In 2020 IEEE International Conference on Image Processing (ICIP) (pp. 370-374). IEEE.

[3] Zhou, T., Li, Q., Lu, H., Cheng, Q., & Zhang, X. (2023). GAN review: Models and medical image fusion applications. Information Fusion, 91, 134-148.

[4] Mi, J., Wang, L., Liu, Y., & Zhang, J. (2022). KDE-GAN: A multimodal medical image-fusion model based on knowledge distillation and explainable AI modules. Computers in Biology and Medicine, 151, 106273.

[5] Rao, V. M., Hla, M., Moor, M., Adithan, S., Kwak, S., Topol, E. J., & Rajpurkar, P. (2025). Multimodal generative AI for medical image interpretation. Nature, 639(8056), 888-896.

[6] Multimodal medical image dataset. Available at: https://github.com/bsun0802/Zero-Learning-Fast-Medical-Image-Fusion/tree/master/images/MRI-PET (Access date: 01-10-2024)

[7] Multimodal medical image dataset. Available at: https://github.com/ashna111/multimodal-image-fusion-to-detect-brain-tumors/tree/master/dataset. (Access date: 13-09-2024)

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